

QRM II Graded Assignment (15)

Period 1, 2026

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0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th**. You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

1 Week 1

1. Find the dataset “movies1.tsv” on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject? [4 points]

```
#WRITE YOUR CODE HERE code
```

```
nrow(movies_one)
```

```
## [1] 505
```

```
#observations  
ncol(movies_one)
```

```
## [1] 19
```

```
# variables  
n_distinct(movies_one$title)
```

```
## [1] 505
```

```
#unique movies (subjects)  
glimpse(movies_one)
```

```
## Rows: 505  
## Columns: 19  
## $ index          <dbl> 1773, 2540, 1174, 3262, 4324, 214, 1737, 610, 2738~  
## $ budget         <dbl> 2.70e+07, 1.50e+07, 4.00e+07, 0.00e+00, 0.00e+00, ~  
## $ keywords       <chr> "fbi island serial killer series of murders", "fir~  
## $ original_language <chr> "en", "en", "en", "en", "en", "en", "en", "en", "e~  
## $ title          <chr> "Mindhunters", "Krampus", "Ride Along 2", "Enough ~  
## $ popularity     <dbl> 17.193659, 31.565117, 25.136082, 14.969093, 0.1478~  
## $ release_date   <date> 2004-05-07, 2015-11-26, 2016-01-14, 2013-09-18, 2~  
## $ revenue        <dbl> 21148829, 61548707, 124827316, 25288872, 0, 325756~
```

```
## $ runtime          <dbl> 106, 98, 102, 93, 97, 130, 108, 99, 100, 99, 82, 1~
## $ vote_average     <dbl> 6.3, 5.9, 6.1, 6.6, 6.0, 6.2, 5.5, 4.4, 6.0, 6.2, ~
## $ vote_count       <dbl> 333, 584, 555, 348, 9, 597, 640, 533, 210, 371, 11~
## $ genre            <chr> "Thriller", "Comedy", "Comedy", "Drama", "Drama", ~
## $ release_year     <dbl> 2004, 2015, 2016, 2013, 2006, 2000, 2016, 2014, 20~
## $ release_month    <dbl> 5, 11, 1, 9, 10, 3, 2, 1, 2, 9, 9, 12, 12, 11, 5, ~
## $ release_day      <dbl> 7, 26, 14, 18, 27, 15, 4, 10, 14, 29, 22, 19, 13, ~
## $ first_actor      <chr> "LL Cool", "Adam Scott", "Kevin Hart", "Julia Loui~
## $ first_actor_gender <chr> NA, "male", "male", "female", "male", "male", "fem~
## $ director_first_name <chr> "Renny", "Michael", "Tim", "Nicole", "Regardt", "W~
## $ director_gender  <chr> "male", "male", "male", "female", NA, "male", "mal~
```

```
#all variables
```

Your Answer:

The dataset contains 505 observations and 19 variables. There are 505 subjects, because every row is a movie and each movie title appears once so the number of subjects equals the number of observations. A subject is a single movie described by variables such as its title, budget, revenue, genre, runtime, release date, ratings and the first actor and director.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); continuous; (iv) discrete. If present, name one example of such a variable present in your data set. [4 points]

```
#WRITE YOUR CODE HERE
glimpse(movies_one)
```

```
# overview of variable types
```

```
## Rows: 505
## Columns: 19
## $ index          <dbl> 1773, 2540, 1174, 3262, 4324, 214, 1737, 610, 2738~
## $ budget         <dbl> 2.70e+07, 1.50e+07, 4.00e+07, 0.00e+00, 0.00e+00, ~
## $ keywords       <chr> "fbi island serial killer series of murders", "fir~
## $ original_language <chr> "en", "en", "en", "en", "en", "en", "en", "en", "e~
## $ title          <chr> "Mindhunters", "Krampus", "Ride Along 2", "Enough ~
## $ popularity     <dbl> 17.193659, 31.565117, 25.136082, 14.969093, 0.1478~
## $ release_date   <date> 2004-05-07, 2015-11-26, 2016-01-14, 2013-09-18, 2~
## $ revenue        <dbl> 21148829, 61548707, 124827316, 25288872, 0, 325756~
## $ runtime        <dbl> 106, 98, 102, 93, 97, 130, 108, 99, 100, 99, 82, 1~
## $ vote_average   <dbl> 6.3, 5.9, 6.1, 6.6, 6.0, 6.2, 5.5, 4.4, 6.0, 6.2, ~
## $ vote_count     <dbl> 333, 584, 555, 348, 9, 597, 640, 533, 210, 371, 11~
## $ genre          <chr> "Thriller", "Comedy", "Comedy", "Drama", "Drama", ~
## $ release_year   <dbl> 2004, 2015, 2016, 2013, 2006, 2000, 2016, 2014, 20~
## $ release_month  <dbl> 5, 11, 1, 9, 10, 3, 2, 1, 2, 9, 9, 12, 12, 11, 5, ~
## $ release_day    <dbl> 7, 26, 14, 18, 27, 15, 4, 10, 14, 29, 22, 19, 13, ~
## $ first_actor    <chr> "LL Cool", "Adam Scott", "Kevin Hart", "Julia Loui~
## $ first_actor_gender <chr> NA, "male", "male", "female", "male", "male", "fem~
## $ director_first_name <chr> "Renny", "Michael", "Tim", "Nicole", "Regardt", "W~
## $ director_gender <chr> "male", "male", "male", "female", NA, "male", "mal~
```

```
unique(movies_one$genre)
```

```
# nominal: categories without order
```

```
## [1] "Thriller"    "Comedy"      "Drama"       "Action"      "Documentary"
## [6] "Family"      NA            "Adventure"   "Fantasy"     "Fiction"
## [11] "Horror"      "Crime"       "Romance"     "Music"
```

```
summary(movies_one$popularity)      # continuous: decimal values
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
## 2.386e-03 4.759e+00 1.414e+01 2.283e+01 2.985e+01 2.037e+02
```

```
summary(movies_one$vote_count)      # discrete: whole numbers (counts)
```

```
##      Min. 1st Qu.  Median     Mean 3rd Qu.    Max.
##      0.0   47.0   255.0   802.9   914.0  11800.0
```

Your Answer: (i) Nominal: A nominal variable has categories without an order. Example genre: Thriller, Comedy, Drama, Action. Other examples are original_language and director_gender. (ii) Ordinal: An ordinal variable has categories with a meaningful order, but no known distance between them like low, medium, high. None of the variables in this data set is an ordered category. (iii) Continuous: Present. A continuous variable can take any value within a range including decimals. Example: popularity. Other examples are revenue, budget and run time. (iv) Discrete: A discrete variable can only take separate, countable values. Example: vote_count, the number of votes a movie received. release_month (1–12) and release_year are other examples.

3. A movie studio wants to know which types of movies give maximal revenue. Perform the following steps to provide the movie studio with an analysis which corresponds to their request:
 - a. Summarize the variable revenue and report its mean, median, maximum, and minimum. [2 points]
 - b. Which movie has the highest revenue in your data set and how much is this revenue. Which movie has the lowest and how much is its revenue? If multiple movies have the exact same highest or lowest revenue, give only one example. [2 points]
 - c. Create a histogram of the variable revenue. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Describe the shape of the histogram. What does this tell you about the nature of making money in the movies industry? [2 points]
 - d. Add a new variable to your data set: the log of revenue. When creating this variable, what happens to movies for which revenue is zero? What then happens when you calculate the mean of log revenue? [2 points]
 - e. For movies that have a revenue of zero, replace log of revenue with “NA”. What is now the mean of log of revenue? Create a histogram for log revenue, again with an appropriate title, x- and y-axis labels. Describe the shape of this histogram. [2 points]
 - f. For the variables revenue and log of revenue calculate their skewness and kurtosis. How do these numbers relate to the shape of the histograms that you estimated under d.) and e.)? [2 points]

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

Step a

```
summary(movies_one$revenue)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## 0.000e+00 0.000e+00 1.948e+07 9.482e+07 1.043e+08 2.788e+09
```

```
mean(movies_one$revenue)
```

```
## [1] 94815482
```

```
median(movies_one$revenue)
```

```
## [1] 19480739
```

```
max(movies_one$revenue)
```

```
## [1] 2787965087
```

```
min(movies_one$revenue)
```

```
## [1] 0
```

Your Answer: The mean revenue is about 94.8 million dollar (94.815.482). The median is about \$19.5 million dollar (19.480.739). The maximum is 2.787.965.087 dollar and the minimum is 0 dollar. The mean is five times higher than the median which shows that a small number of successful movies increase average.

Step b

```
#WRITE YOUR CODE HERE
```

```
movies_one[which.max(movies_one$revenue),
            c("title", "revenue")]
```

```
## # A tibble: 1 x 2
##   title      revenue
##   <chr>         <dbl>
## 1 Avatar 2787965087
```

```
movies_one[which.min(movies_one$revenue),
            c("title", "revenue")]
```

```
## # A tibble: 1 x 2
##   title      revenue
##   <chr>         <dbl>
## 1 Faith Like Potatoes      0
```

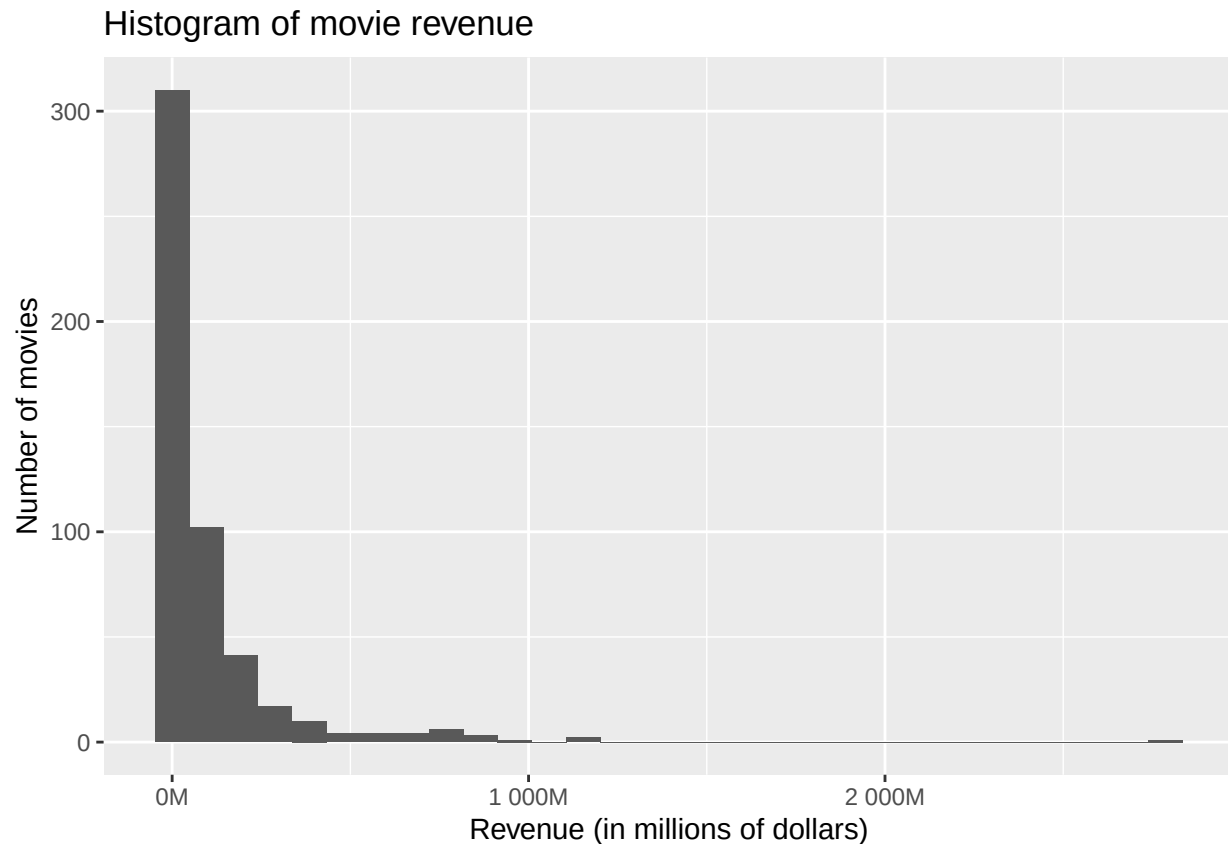
```
sum(movies_one$revenue == 0)
```

```
## [1] 160
```

Your Answer: The movie with the highest revenue is Avatar, with a revenue of 2,787,965,087 dollar. The lowest revenue is 0 dollar, which is shared by 160 movies; one example is Faith Like Potatoes.

Step c

```
#WRITE YOUR CODE HERE
ggplot(movies_one, aes(x = revenue)) +
  geom_histogram(bins = 30) +
  scale_x_continuous(labels = scales::label_number(scale = 1e-6, suffix = "M")) +
  labs(title = "Histogram of movie revenue",
       x = "Revenue (in millions of dollars)",
       y = "Number of movies")
```



Your Answer: The Histogram show that a lot of movies don't make a lot of money. It is very right sided. Ninety percent of the movies earn less than 244 million dollars while the highest revenue is almost 2.8 billion dollars. This shows that making money in the movie industry is very concentrated with a few movies. Most movies earn little or nothing while a few blockbusters earn extremely high revenues.

Step d

```
#WRITE YOUR CODE HERE
movies_one$log_revenue <- log(movies_one$revenue)

sum(movies_one$revenue == 0)
```

```
## [1] 160
```

```
mean(movies_one$log_revenue)
```

```
## [1] -Inf
```

Your Answer:

For movies with a revenue of zero, the log is undefined. Rstudio returns -Inf (minus infinity), since $\log(0) = -\infty$. This affects 160 movies. Because of this the mean of log revenue is -Inf and is therefore not useful.

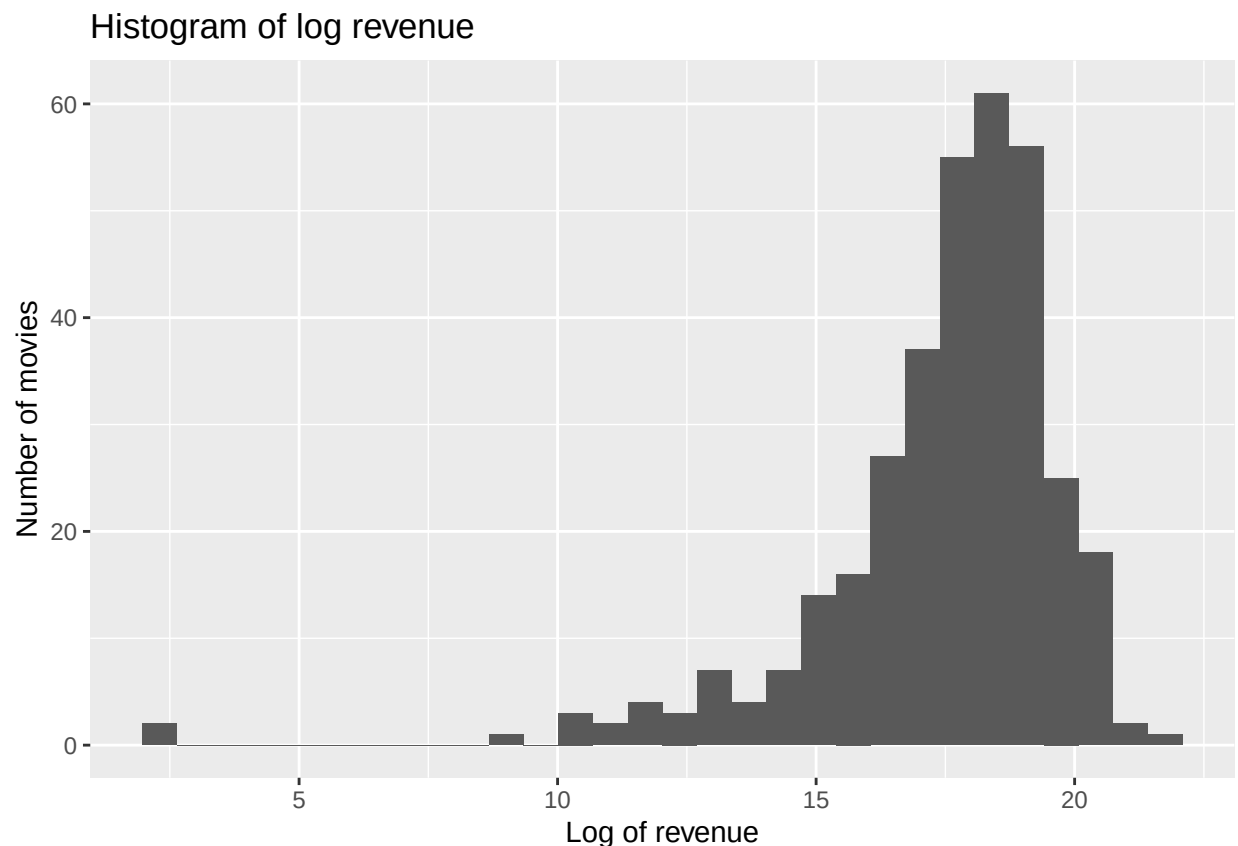
Step e

```
#WRITE YOUR CODE HERE
movies_one$log_revenue <- ifelse(movies_one$revenue == 0, NA, log(movies_one$revenue))

mean(movies_one$log_revenue, na.rm = TRUE)
```

```
## [1] 17.4498
```

```
ggplot(movies_one, aes(x = log_revenue)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(title = "Histogram of log revenue",
       x = "Log of revenue",
       y = "Number of movies")
```



Your Answer: When replacing the log of zero revenue with N.A. the mean of log revenue is about 17.45 using the left over 345 movies. The histogram is uni modal and left-skewed. Most values lie between roughly 16 and 21, with a peak around 18, and there is a thin tail to the left formed by a few movies with a very low positive revenue. The shape is much more symmetric than the histogram of the raw revenue.

Step f

```
#WRITE YOUR CODE HERE
```

```
skewness(movies_one$revenue)
```

```
## [1] 6.049774
```

```
kurtosis(movies_one$revenue)
```

```
## [1] 63.28088
```

```
skewness(movies_one$log_revenue, na.rm = TRUE)
```

```
## [1] -2.290049
```

```
kurtosis(movies_one$log_revenue, na.rm = TRUE)
```

```
## [1] 12.5771
```

Your Answer:

For revenue the skewness is about 6,05 and the kurtosis is about 63,3. For log revenue the skewness is about -2,29 and the kurtosis is about 12,6. A skewness of 6,05 confirms the strong right skew that we see in the histogram of revenue, and a kurtosis far above 3 the value for a normal distribution means heavy tails so extreme outliers such as Avatar. For log revenue the skewness is negative in line with the thin left tail in the histogram and both numbers are much closer to those of a normal distribution. The kurtosis of 12,6 is still above 3, so the log transformation makes the distribution more symmetric but does not make it fully normal.

2 Week 2

1 Is your dataset movies1.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? **[4 points]**

Your answer here:

The dataset movies1.tsv (movie_one) is a sample not the full population. It contains only 505 movies, while there are thousands of movies released yearly, so this is clearly a smaller part rather than every movie that exists.

2 For this question, you will assume that your data set is the full population.

- a. What is the mean of the variable vote_average in your data set? **[2 points]**
- b. Create a new data set, called movies_sample. Make sure that it is a random sample of your data set of 25 movies. What is the mean of vote_average in this random sample? How does it compare to the mean of vote_average in 2a? **[2 points]**
- c. In a for loop, create 100 different samples of 25 movies, as in b, and estimate the mean within each sample. Save the mean of each sample in a vector called sample_means. So the first position of the vector would have the mean of the first sample, the second position the mean of the second sample, etc. Print the start of this vector. **[2 points]**

- d. Summarize and make a histogram of `sample_means`. What is the mean, standard deviation and shape of its distribution? [2 points]
- e. Using the formulas of the *Central Limit Theorem*, give the distribution of the sample mean of `vote_average` in a sample of 25 movies from your population (including its expected value and standard deviation). How do these values compare to your answers at d.)? [2 points]

Step a

```
#WRITE YOUR CODE HERE
mean(movies_one$vote_average)
```

```
## [1] 6.042772
```

Your Answer:

The mean of `vote_average` in the full population (505 movies) is 6.042772 Step b

```
#WRITE YOUR CODE HERE
set.seed(123)
movies_sample <- movies_one[sample(nrow(movies_one), 25), ]
mean(movies_sample$vote_average)
```

```
## [1] 5.904
```

Your Answer:

The mean of `vote_average` in this random sample of 25 movies is close (5.904) but not exactly equal to the population mean of 6.04 from the question before. Because the sample only contains 25 out of 505 its mean is affected by sampling variation: a different random sample would give a slightly different mean. Still, since the sample was drawn randomly it should on average be a reasonable estimate of the population mean

Step c

```
#WRITE YOUR CODE HERE
set.seed(123)
sample_means <- numeric(100)
for (i in 1:100) {
  sample_i <- movies_one[sample(nrow(movies_one), 25), ]
  sample_means[i] <- mean(sample_i$vote_average)
}
head(sample_means)
```

```
## [1] 5.904 5.528 6.096 6.020 6.168 5.824
```

Your Answer:

The vector `sample_means` contains the mean `vote_average` of 100 different random samples of 25 movies. The first values of the vector are printed above with `head()`.

Step d

```
#WRITE YOUR CODE HERE
```

```
summary(sample_means)
```

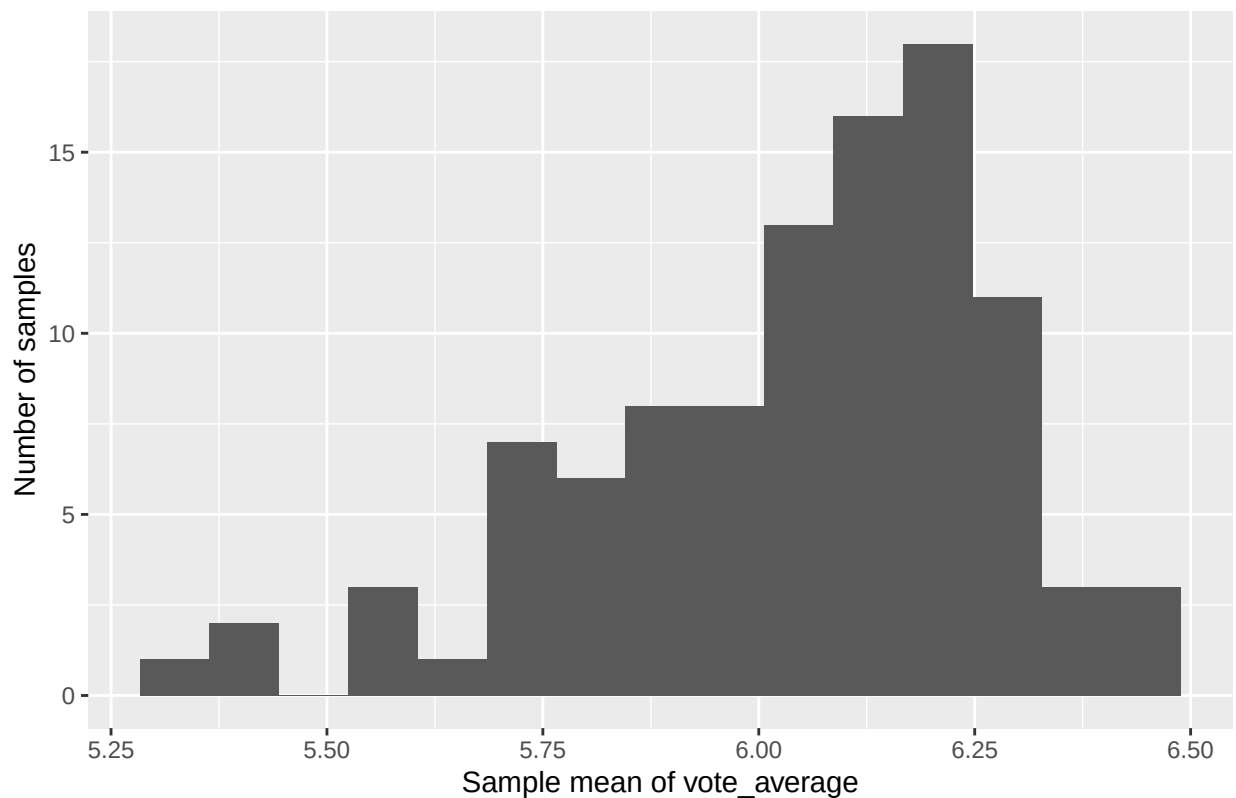
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  5.324   5.916   6.096   6.037   6.196   6.448
```

```
sd(sample_means)
```

```
## [1] 0.2337529
```

```
ggplot(data.frame(sample_means), aes(x = sample_means)) +  
  geom_histogram(bins = 15) +  
  labs(title = "Histogram of sample means of vote_average (100 samples of n = 25)",  
        x = "Sample mean of vote_average",  
        y = "Number of samples")
```

Histogram of sample means of vote_average (100 samples of n = 25)



Your Answer:

The mean of sample_means is approximately 6.037 and is very close to the population mean of 6.04. The standard deviation of sample_means is approximately 0.2337529. The histogram is bell-shaped centered around a sample mean of roughly 6.037, with a slight left skew. This shape is close to what the Central Limit Theorem predicts: even though only 100 samples were drawn, the distribution of sample means already resembles a normal distribution.

Step e

```
pop_mean <- mean(movies_one$vote_average)
pop_sd <- sd(movies_one$vote_average)
```

```
pop_mean                                # expected value of the sample mean
```

```
## [1] 6.042772
```

```
pop_sd / sqrt(25)                       # standard deviation (standard error) of the sample mean
```

```
## [1] 0.2383468
```

Your Answer:

According to the Central Limit Theorem, for a large enough sample size the distribution of the sample mean is approximately normal, ...with expected value equal to the population mean, $\mu = 6.043$, and standard deviation (standard error) equal to $\sigma/\sqrt{n} = 1.192/\sqrt{25} \approx 0.238$. This is very close to what we found by simulation in d.: a mean of 6.096 and a standard deviation of 0.2337529. The small differences are due to sampling variation, since we only drew 100 samples rather than an infinite number of samples. This confirms the Central Limit Theorem: even though the original variable `vote_average` is not perfectly normally distributed, the distribution of its sample mean is approximately normal and centered on the population mean, with a spread that shrinks as the sample size n increases.

3. Is there a variable in your data set which you would expect to follow a discrete uniform distribution? Explain, without looking at the data, why you expect this data to be discrete uniform. If the data would indeed follow a discrete uniform distribution, what would you expect the mean and standard deviation of this variable to be? Next, plot a histogram of the variable, and plot an appropriate discrete uniform distribution on top of this histogram. Comment whether the variable indeed follows this expected distribution. If the distribution deviates from the discrete uniform distribution, explain why this is the case. **[6 points]**

Your Answer:

we expect `release_month` (the month in which a movie is released, 1–12) to follow a discrete uniform distribution. Without looking at the data, there is no a priori reason why movies would be more likely to release in one particular month than another: a movie can in principle be released in any of the 12 months, and each month is equally likely if release timing were purely random. For a discrete uniform distribution on the integers 1 to 12, the expected mean is $E(X) = \frac{1+n}{2} = \frac{1+12}{2} = 6.5$, and the expected standard deviation is $SD(X) = \sqrt{\frac{n^2-1}{12}} = \sqrt{\frac{144-1}{12}} \approx 3.45$.

```
#WRITE YOUR CODE HERE
# Theoretical values for a discrete uniform distribution on {1, 2, ..., 12}
n <- 12
theoretical_mean <- (1 + n) / 2
theoretical_sd <- sqrt((n^2 - 1) / 12)

theoretical_mean
```

```
## [1] 6.5
```

```
theoretical_sd
```

```
## [1] 3.452053
```

```
# Actual mean and sd in the data  
mean(movies_one$release_month)
```

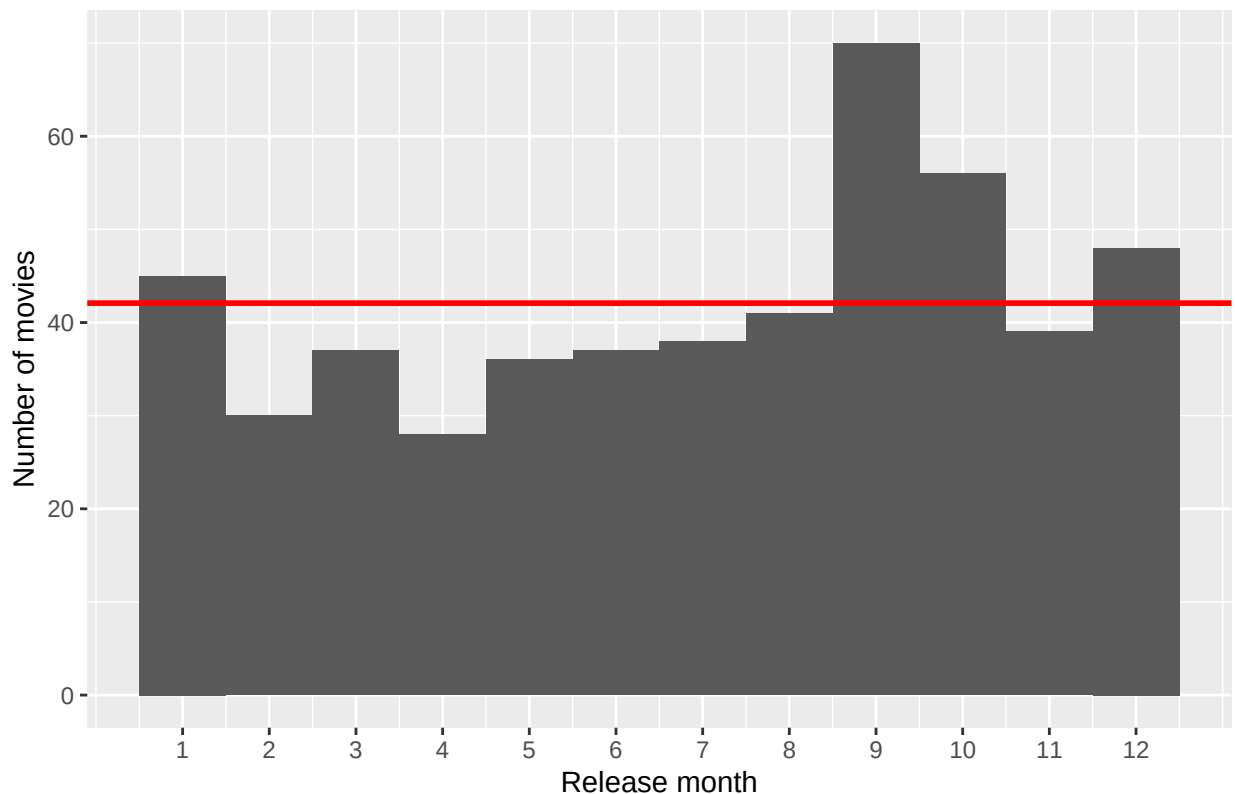
```
## [1] 6.968317
```

```
sd(movies_one$release_month)
```

```
## [1] 3.462811
```

```
# Histogram with the discrete uniform distribution overlaid  
ggplot(movies_one, aes(x = release_month)) +  
  geom_histogram(aes(y = after_stat(count)), binwidth = 1, boundary = 0.5) +  
  geom_hline(yintercept = nrow(movies_one) / n, color = "red", linewidth = 1) +  
  scale_x_continuous(breaks = 1:12) +  
  labs(title = "Histogram of release month vs. discrete uniform distribution",  
        x = "Release month",  
        y = "Number of movies")
```

Histogram of release month vs. discrete uniform distribution



Your Answer:

I expect `release_month` (1–12) to follow a discrete uniform distribution, since there is no a priori reason a movie would be more likely to release in one month than another. For a discrete uniform distribution on $\{1, \dots, 12\}$, the expected mean is $\frac{1+12}{2} = 6.5$ and the expected standard deviation is $\sqrt{\frac{12^2-1}{12}} \approx 3.45$. The observed mean (6.97) and standard deviation (3.46) are close to these theoretical values, but the histogram shows the distribution is not truly uniform: there is a clear peak around September–October and relatively low counts in February and April. This deviates from the flat pattern expected under a discrete uniform distribution. This makes sense because release timing is not random: studios strategically schedule releases around the autumn festival season, while months like February and April are typically considered weaker box-office periods and are avoided for major releases.

3 Week 3

For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.

1. Give a 90 percent confidence interval for the variable *revenue*. Interpret the result. [2 points]

#WRITE YOUR CODE HERE

```
n <- length(movies_one$revenue)
mean_revenue <- mean(movies_one$revenue)
sd_revenue <- sd(movies_one$revenue)
se_revenue <- sd_revenue / sqrt(n)

t_crit <- qt(0.95, df = n - 1)

lower <- mean_revenue - t_crit * se_revenue
upper <- mean_revenue + t_crit * se_revenue

lower
```

```
## [1] 79639410
```

```
upper
```

```
## [1] 109991553
```

Your Answer:

The 90% confidence interval for the mean revenue is approximately [79,639,410 ; 109,991,553]. This means that if we were to repeat this sampling many times and construct a 90% confidence interval each time, about 90% of those intervals would contain the real population mean of the revenue. We are 90% confident that the true mean revenue of the full population of movies lies between roughly 79.6 and 110.0 million dollar.

2

- a. How many movies in your sample were directed by a male director? How many by a female director? [2 points]

- b. Test the null hypothesis that the chance that a movie is being directed by a male or female director is equal. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Use an appropriate R function to test your hypothesis, and state your conclusion. Also give the p-value that corresponds to your test statistic. **[2 points]**
- c. Recode revenue such that it is measured in millions of dollars. What is the variance of revenue for movies with a male director? What is the variance of revenue for movies with a female director? **[2 points]**
- d. Assume that the variance of revenue for movies with a male director in your data is the same as the variance for movies with a male director in your population (that is, there is no sampling uncertainty in this estimate). Now, set up a hypothesis test to test for the null hypothesis that the variance of revenue for movies with a female director is equal to the variance for movies with a male director. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Calculate your test statistic and tell how your test statistic is distributed, and state your conclusion. Also give the p-value that corresponds to your test statistic. **[2 points]**

step a

#WRITE YOUR CODE HERE

```
table(movies_one$director_gender)
```

```
##
## female    male
##      44    426
```

```
sum(movies_one$director_gender == "male", na.rm = TRUE)
```

```
## [1] 426
```

```
sum(movies_one$director_gender == "female", na.rm = TRUE)
```

```
## [1] 44
```

Your Answer:

In the sample, 426 movies were directed by a male director and 44 movies by a female director. There are also 35 movies with a missing (NA) value for director_gender

step b

#WRITE YOUR CODE HERE

```
binom.test(x = 426, n = 426 + 44, p = 0.5)
```

```
##
## Exact binomial test
##
## data: 426 and 426 + 44
## number of successes = 426, number of trials = 470, p-value < 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
```

```
## 0.8763630 0.9311527
## sample estimates:
## probability of success
## 0.906383
```

Your Answer:

Null hypothesis H_0 : the probability that a movie is directed by a male director is $P = 0.5$ (male and female directors are equally likely). **Alternative hypothesis H_1 :** $p \neq 0.5$. Significance level: $\alpha = 0.05$. Of the 470 movies with a known director gender, 426 were directed by a man. A two-sided binomial test against $p = 0.5$ gives a test statistic based on 426 successes out of 470 trials, with a p-value that is essentially 0 (far below 0.001). Because the p-value is much smaller than $\alpha = 0.05$, we reject the null hypothesis. There is very strong evidence that male and female directors are not equally likely in this population; movies are directed by men far more often than by women.

step c

```
#WRITE YOUR CODE HERE

movies_one$revenue_millions <- movies_one$revenue / 1e6

var(movies_one$revenue_millions[movies_one$director_gender == "male"], na.rm = TRUE)

## [1] 48054.81
```

```
var(movies_one$revenue_millions[movies_one$director_gender == "female"], na.rm = TRUE)

## [1] 17490.31
```

Your Answer:

After recording revenue into millions of dollars, the variance of revenue for movies with a male director is approximately 48,055 (million squared), and for movies with a female director it is approximately 17,490 (million squared).

step d

```
#WRITE YOUR CODE HERE

sigma_sq_male <- var(movies_one$revenue_millions[movies_one$director_gender == "male"],
  ↪ na.rm = TRUE) # treated as known population variance
s_sq_female <- var(movies_one$revenue_millions[movies_one$director_gender == "female"],
  ↪ na.rm = TRUE)
n_female <- sum(movies_one$director_gender == "female", na.rm = TRUE)

chi_sq_stat <- (n_female - 1) * s_sq_female / sigma_sq_male
chi_sq_stat

## [1] 15.65053
```

```
df <- n_female - 1
df
```

```
## [1] 43
```

```
p_value <- 2 * min(pchisq(chi_sq_stat, df), 1 - pchisq(chi_sq_stat, df))
p_value
```

```
## [1] 8.252287e-05
```

Your Answer:

H0: $\sigma_{female}^2 = \sigma_{male}^2$ (approximately 48,055, treated as a known population value). $H_1: \sigma_{female}^2 \neq \sigma_{male}^2$. $\alpha = 0.05$. Since σ_{male}^2 is known, we test s_{female}^2 against it using $\chi^2 = \frac{(n-1)s_{female}^2}{\sigma_{male}^2}$, which follows a chi-square distribution with $n-1 = 43$ degrees of freedom under H_0 . The test statistic is $\chi^2 \approx 15.65$, giving a two-sided p-value of approximately 0.00008. Since this is well below $\alpha = 0.05$, we reject H0: the variance of revenue for female-directed movies differs from (and is lower than) that for male-directed movies.

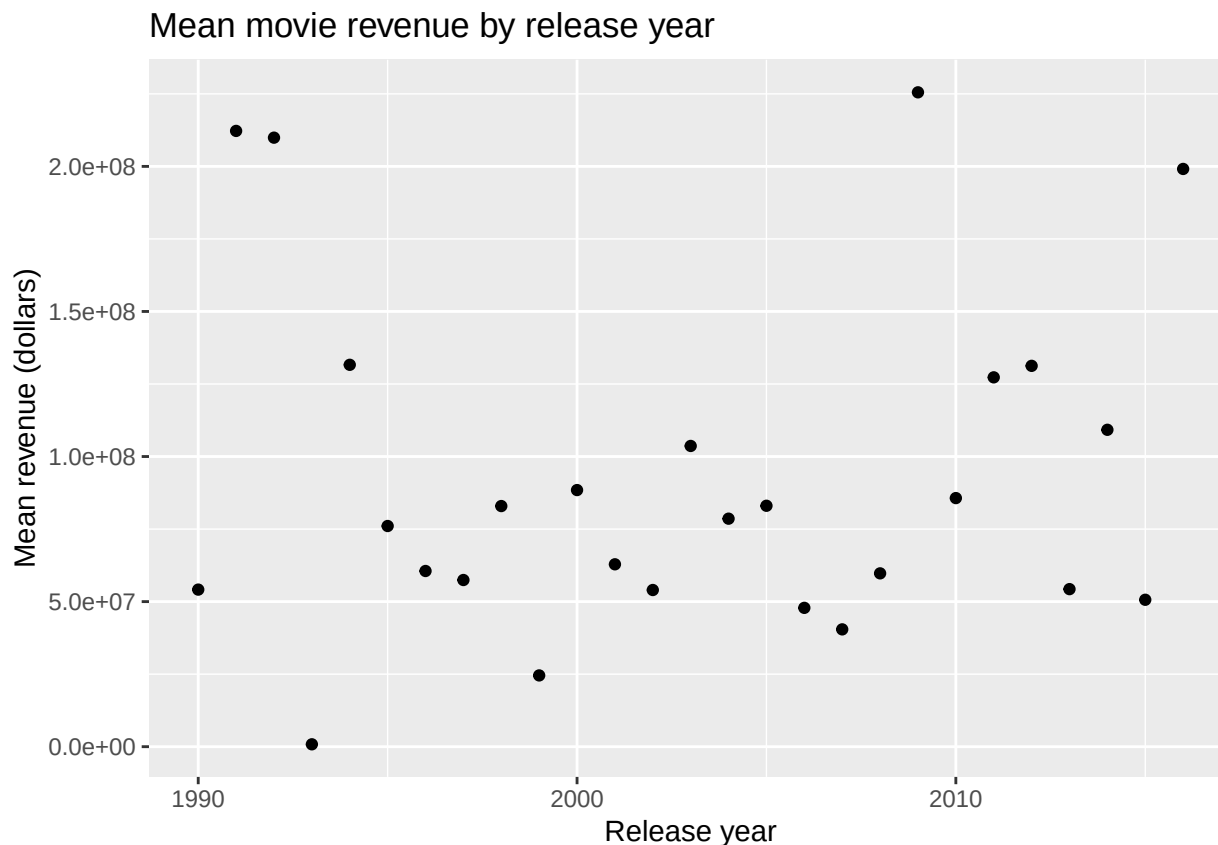
3

- Create a scatter plot with year on the x axis and mean of revenue within that year on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Is revenue of movies increasing or decreasing over time? **[3 points]**
- Recreate the scatter plot with year on the x axis and mean_revenue on the y-axis, but now add bars around each point, indicating the 95% confidence interval. **[3 points]**
- If you look closely at the resulting plot in b.), you will probably see that the confidence intervals in some years are wider than in other years. What are some possible reasons that a confidence interval in a given year can be wide? Can you verify this using your data? **[4 points]**

step a

```
#WRITE YOUR CODE HERE
yearly_revenue <- movies_one %>%
  group_by(release_year) %>%
  summarize(mean_revenue = mean(revenue))

ggplot(yearly_revenue, aes(x = release_year, y = mean_revenue)) +
  geom_point() +
  labs(title = "Mean movie revenue by release year",
       x = "Release year",
       y = "Mean revenue (dollars)")
```

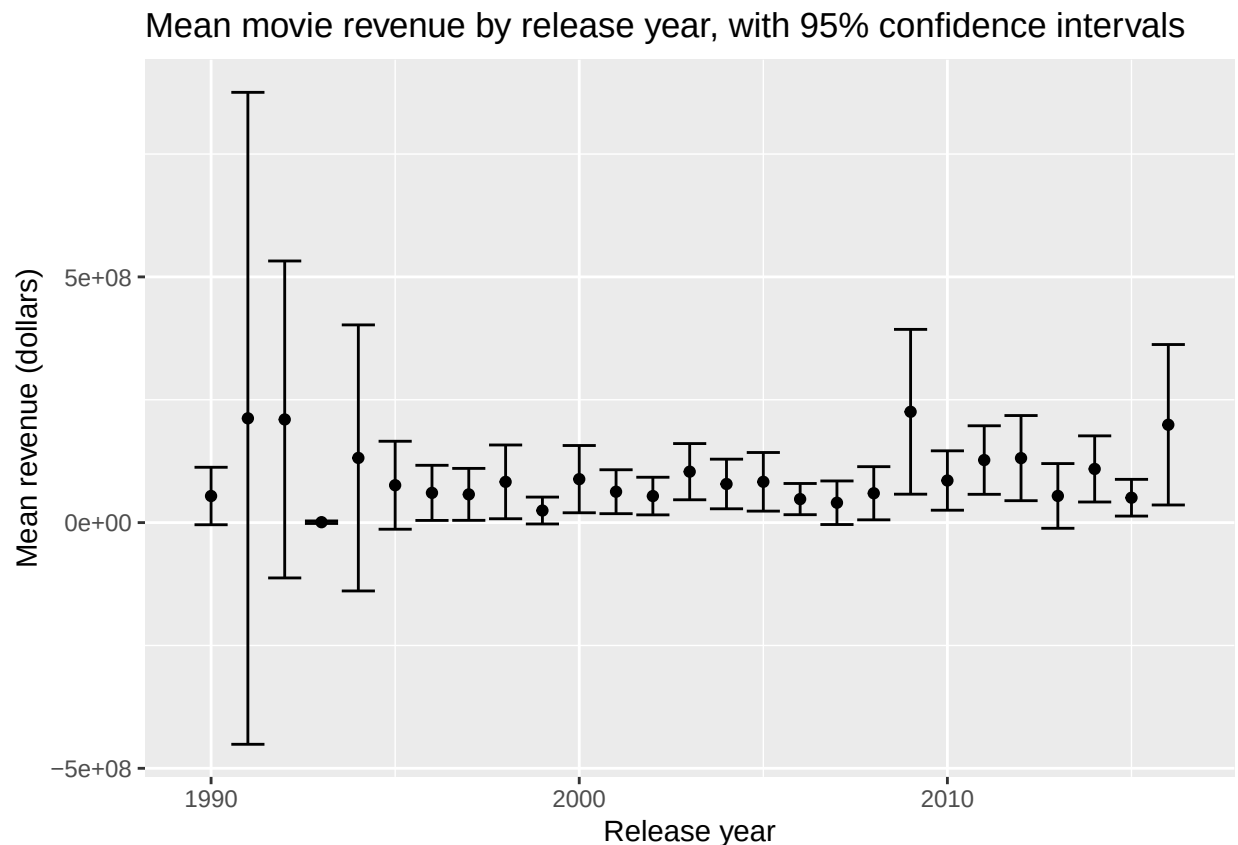


Your Answer:

The scatter plot does not show a clear upward or downward trend in mean revenue over time. There is a lot of variation each year with some years showing very high average revenue and others much lower but no strong overall pattern. *step b*

```
#WRITE YOUR CODE HERE
yearly_revenue <- movies_one %>%
  group_by(release_year) %>%
  summarize(mean_revenue = mean(revenue),
            sd_revenue = sd(revenue),
            n = n(),
            se = sd_revenue / sqrt(n),
            t_crit = qt(0.975, df = n - 1),
            ci_lower = mean_revenue - t_crit * se,
            ci_upper = mean_revenue + t_crit * se)

ggplot(yearly_revenue, aes(x = release_year, y = mean_revenue)) +
  geom_point() +
  geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper)) +
  labs(title = "Mean movie revenue by release year, with 95% confidence intervals",
       x = "Release year",
       y = "Mean revenue (dollars)")
```



Your Answer:

The plot now shows the mean revenue per year together with a 95% confidence interval around each point, illustrating how precisely the mean is estimated for each year.

step c

```
#WRITE YOUR CODE HERE
yearly_revenue %>%
  arrange(desc(ci_upper - ci_lower)) %>%
  select(release_year, n, sd_revenue, ci_lower, ci_upper)
```

```
## # A tibble: 27 x 5
##   release_year     n sd_revenue  ci_lower  ci_upper
##   <dbl> <int>   <dbl>   <dbl>   <dbl>
## 1      1991     3 267045531. -451150178. 875605571.
## 2      1992     3 129834690. -112623364. 532431134.
## 3      1994     7 292685338. -139072255. 402305326.
## 4      2009    37 502939570.   57838647. 393215449.
## 5      2016    19 338706404.   35872184. 362374706.
## 6      1995     9 116438509.  -13443845. 165561277.
## 7      2012    36 256066373.   44614323. 217895114.
## 8      1998    15 135533982.    7875376. 157987811.
## 9      2011    22 157312285.   57549114. 197045834.
## 10     2000    19 141945943.   20026655. 156858164.
## # i 17 more rows
```

Your Answer:

A confidence interval for a given year can be wide for two main reasons: (1) a small sample size in that year, since the standard error is $s/\sqrt{n}$ and shrinks as n grows, and (2) a large standard deviation of revenue within that year, meaning revenue values are very spread out (e.g. a mix of blockbusters and low-revenue movies). This can indeed be verified in the data. For example, 1991 and 1992 have only 3 movies each, and their confidence intervals are among the widest in the plot, showing that a small sample size drives up the width. Similarly, 2009 has a much larger sample (37 movies) but still has one of the widest intervals, because its standard deviation of revenue is very high (movies that year ranged from low-revenue films to a very high-grossing blockbuster). In contrast, years like 1993 have both a small n and a low standard deviation, resulting in a narrow interval. So both a small sample size and high within-year variability contribute to wider confidence intervals, and the data confirms that both factors are present in different years.

4 Week 4

1. There is an argument going on in the movie studio. *Bob* claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. *Chantal* disagrees. She tells Bob that “Every dollar we spend on movie production is more than offset by the increase in movie revenue”.

- a. Set up a regression model to test Chantal’s claim, and estimate it. That is, estimate:

$$\text{Revenue}_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i.$$

Print a summary of your estimated model. **[2 points]**

- b. What is the estimated value of β_1 and how do you interpret it? **[2 points]**
- c. Test for the null hypothesis that $\beta_1 \leq 1$. Report the p-value and state your conclusion. **[2 points]**
- d. Next, estimate the model

$$\text{Log Revenue}_i = \beta_0 + \beta_1 \text{Log Budget}_i + \varepsilon_i.$$

When creating the variables Log Revenue and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value “NA”. Print a summary of your estimated model. **[2 points]**

- e. What is the estimated value of β_1 and how do you interpret it? **[2 points]**
- f. Which model has better fit? The level-level model or the log-log model? Explain. **[2 points]**
- g. Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? **[2 points]**

step a

```
#WRITE YOUR CODE HERE
model_level <- lm(revenue ~ budget, data = movies_one)
summary(model_level)
```

```
##
## Call:
## lm(formula = revenue ~ budget, data = movies_one)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -420678243  -37658344   14833690   17600279  1964947903
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -1.760e+07  7.165e+06  -2.456   0.0144 *
## budget      3.547e+00  1.302e-01  27.248  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 131700000 on 503 degrees of freedom
## Multiple R-squared:  0.5961, Adjusted R-squared:  0.5953
## F-statistic: 742.5 on 1 and 503 DF,  p-value: < 2.2e-16
```

The estimated level-level regression model is $\widehat{\text{Revenue}}_i = -17,600,000 + 3.547 \times \text{Budget}_i$.

step b

```
#WRITE YOUR CODE HERE
coef(model_level)["budget"]
```

```
##    budget
## 3.546909
```

The estimated slope coefficient is β_1 is approximately 3.547: on average, each extra dollar of production budget is associated with about \$3.55 more revenue, holding all else constant. This effect is statistically significant ($p < 0.001$).

step c

```
#WRITE YOUR CODE HERE
beta1 <- coef(model_level)["budget"]
se1 <- summary(model_level)$coefficients["budget", "Std. Error"]
df_resid <- model_level$df.residual

t_stat <- (beta1 - 1) / se1
t_stat
```

```
##    budget
## 19.56605
```

```
p_value <- 1 - pt(t_stat, df = df_resid)
p_value
```

```
## budget
##      0
```

Null hypothesis (H_0): $\beta_1 \leq 1$. Alternative hypothesis (H_1): $\beta_1 > 1$. Significance level: $\alpha = 0.05$.

The test statistic is $t = \frac{3.547-1}{0.130} \approx 19.57$, with a corresponding (one-sided) p-value that is essentially 0 (far below 0.001). Since the p-value is much smaller than $\alpha = 0.05$, we reject the null hypothesis. There is strong evidence that $\beta_1 > 1$, meaning that a dollar spent on production budget is associated with more than a dollar increase in revenue, which supports Chantal's claim.

step d

```
#WRITE YOUR CODE HERE
movies_one$log_revenue <- ifelse(movies_one$revenue == 0, NA, log(movies_one$revenue))
movies_one$log_budget <- ifelse(movies_one$budget == 0, NA, log(movies_one$budget))

model_log <- lm(log_revenue ~ log_budget, data = movies_one)
summary(model_log)
```

```
##
## Call:
## lm(formula = log_revenue ~ log_budget, data = movies_one)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.7790  -0.5049   0.2156   0.7958   4.9779
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.9906     1.0826   1.839  0.0669 .
## log_budget    0.9207     0.0633  14.545 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.624 on 322 degrees of freedom
## (181 observations deleted due to missingness)
## Multiple R-squared:  0.3965, Adjusted R-squared:  0.3946
## F-statistic: 211.6 on 1 and 322 DF,  p-value: < 2.2e-16
```

The estimated log-log regression model is $\widehat{\text{Log Revenue}}_i = 1.991 + 0.921 \times \text{Log Budget}_i$, estimated on 324 movies (after dropping movies with a revenue or budget of zero).

step e

```
#WRITE YOUR CODE HERE
coef(model_log)["log_budget"]
```

```
## log_budget
## 0.9207402
```

The estimated value of β_1 is approximately 0.921. β_1 can be interpreted as an elasticity: a 1% increase in production budget is associated with an approximately 0.92% increase in revenue, on average. This coefficient is statistically significant ($p < 0.001$).

step f

```
#WRITE YOUR CODE HERE
summary(model_level)$r.squared
```

```
## [1] 0.5961367
```

```
summary(model_log)$r.squared
```

```
## [1] 0.3965097
```

The level-level model has an R^2 of approximately 0.596, while the log-log model has an R^2 of approximately 0.397. Based on R^2 alone, the level-level model appears to fit better. However, this comparison is not entirely fair: the two models are estimated on different numbers of observations (505 for the level-level model vs. 324 for the log-log model, since movies with zero revenue or budget are dropped in the log-log model), and the dependent variables are on different scales (dollars vs. log-dollars), so their R^2 values are not directly comparable in a strict sense. Looking at the diagnostics, the level-level model also shows extremely high skewness and kurtosis in its residuals (skew = 6.99, kurtosis = 103), suggesting its errors are far from normally distributed and likely driven by a few extreme outliers (like Avatar). The log-log model's residuals are much closer to a well-behaved distribution, even though its raw R^2 is lower. This suggests the log-log model may actually provide a more reliable fit, even though it explains less variance on the original scale.

step g

Based on the regression results, Chantal appears to be correct: in both the level-level and log-log models, the relationship between budget and revenue is positive and statistically significant, and the hypothesis test in c. confirms that each dollar of budget is associated with more than a dollar of additional revenue ($\beta_1 \approx 3.55 > 1$). This suggests that, on average, production spending is more than offset by the resulting increase in revenue.

However, we would advise the studio to be cautious before concluding that increasing budgets is always a good strategy. This is a correlational, not necessarily causal, relationship: successful movies may attract larger budgets in anticipation of success (reverse causality), and the result is heavily influenced by a small number of very high-budget blockbusters. We would advise the studio not to simply conclude “always spend more”, but to investigate which types of movies generate the best return per dollar spent, rather than assuming the same relationship holds equally for every production.

2

- Make a scatterplot with log revenue on the y-axis, and log budget on the x-axis. Within the same scatterplot, draw a straight line which summarizes the association between the log of revenue and log of budget. **[3 points]**
- Using the insights from the scatterplot, your linear model, and your data, find an example of a movie that vastly overperformed. That means, find a movie that had much more revenue than expected, given its budget. For this movie, report its log of revenue, its predicted log of revenue, and its residual. **[3 points]**

step a

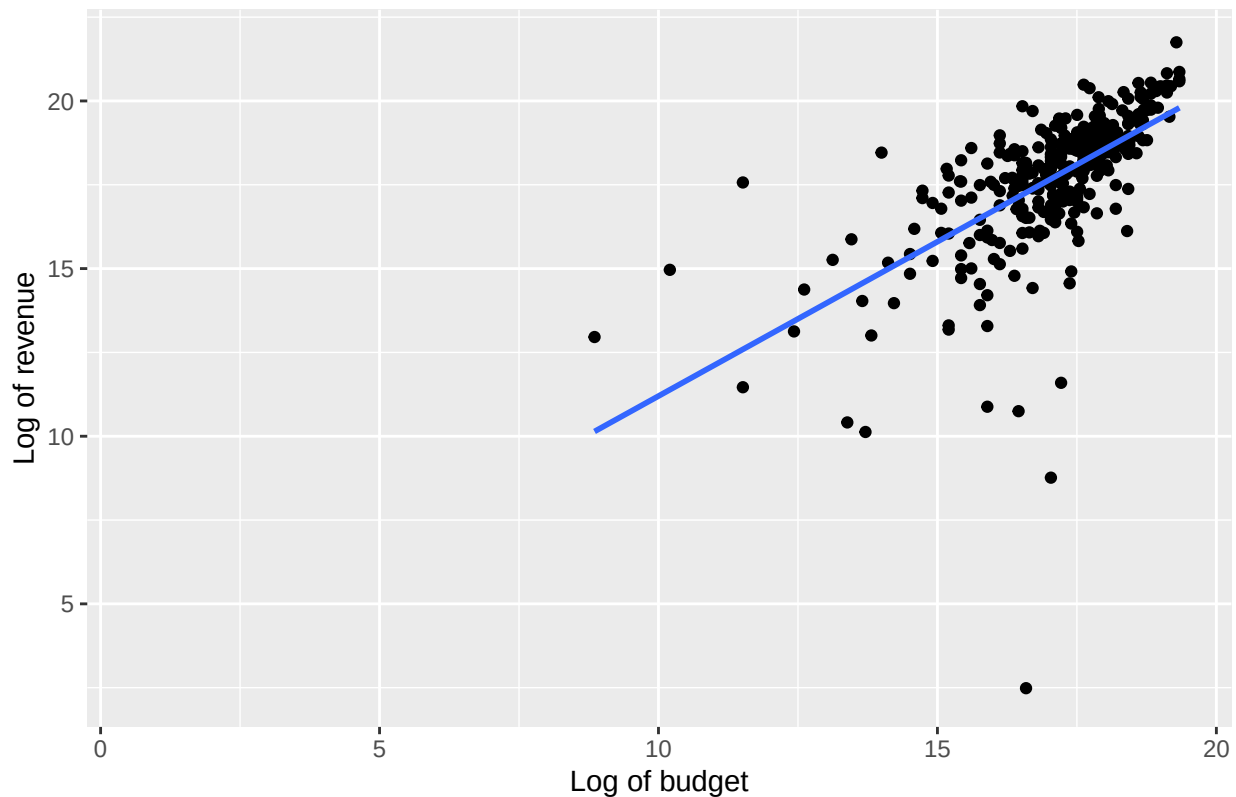
```
#WRITE YOUR CODE HERE
ggplot(movies_one, aes(x = log_budget, y = log_revenue)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Log revenue vs. log budget",
       x = "Log of budget",
       y = "Log of revenue")

## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 181 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 181 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Log revenue vs. log budget



The scatter plot shows a clear positive association between log budget and log revenue: movies with higher budgets tend to also have higher revenue. The fitted straight line, based on the regression model from question 1d, summarizes this relationship. There is still considerable scatter around the line, meaning that budget alone does not perfectly predict revenue.

step b

```
#WRITE YOUR CODE HERE
movies_one$predicted_log_revenue <- predict(model_log, newdata = movies_one)
movies_one$residual <- movies_one$log_revenue - movies_one$predicted_log_revenue

movies_one %>%
  arrange(desc(residual)) %>%
  select(title, log_revenue, predicted_log_revenue, residual) %>%
  head(5)
```

```
## # A tibble: 5 x 4
##   title          log_revenue predicted_log_revenue residual
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 The Gallows    17.6            12.6           4.98
## 2 Saw            18.5            14.9           3.58
## 3 Clerks        15.0            11.4           3.58
## 4 Primer        13.0            10.1           2.82
## 5 The King's Speech 19.8            17.2           2.64
```

A movie that vastly overperformed, given its budget, is “The Gallows”. Its log of revenue was approximately 17.57, while the model predicted a log of revenue of approximately 13.33, given its (low) budget of only

100,000 dollars. This gives a residual of approximately 4.24, the largest positive residual in the data set. In other words, the actual revenue of \$42.7 million was far higher than what the model would expect based on its small budget alone, this movie earned much more than its budget would predict.

5 Week 5

- Make a frequency table of the variable genre. Which are the three genres that occur the most? **[2 points]**
- Create a new variable called genre2. Make sure that the three most popular genres remain the same, but that all the other genres get categorized into one genre called “Other”. Report a frequency table of this new variable genre2. **[2 points]**
- Estimate an OLS model which has as dependent variable the log of revenue of a movie, and as independent variable the log of budget, a dummy for each level that genre2 can take, and the year of release. Show a summary of the resulting model and interpret each coefficient. **[4 points]**

The movie studio that you work at is releasing a new movie in 2026. It will be a Drama movie with a budget of 20,000,000. It will have a runtime of 120 minutes.

- Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 90% prediction interval. **[5 points]**
- Plot the residuals against various independent variables in the model. Do you find any evidence of heteroskedasticity? **[2 points]**
- One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Another executive is of the opinion that the month in which the movie is released will not have a big impact on its revenue. Estimate a model that can test whether the month of release impacts the revenue of a movie. Use an appropriate hypothesis test to test whether month of release matters. What would you advise the movie studio regarding the timing of the release of the movie? **[5 points]**

step a

```
#WRITE YOUR CODE HERE
table(movies_one$genre)
```

```
##
##      Action  Adventure      Comedy      Crime Documentary      Drama
##         38         8        104         2         11         244
##      Family  Fantasy      Fiction      Horror      Music      Romance
##         10         2         6         8         1         1
##      Thriller
##         65
```

```
sort(table(movies_one$genre), decreasing = TRUE)
```

```
##
##      Drama      Comedy      Thriller      Action Documentary      Family
##      244      104      65      38      11      10
## Adventure      Horror      Fiction      Crime      Fantasy      Music
##         8         8         6         2         2         1
##      Romance
##         1
```

The three most common genres are Drama 244 movies, Comedy 104 movies and Thriller 65 movies.

step b

```
#WRITE YOUR CODE HERE
movies_one$genre2 <- ifelse(movies_one$genre %in% c("Drama", "Comedy", "Thriller"),
                           movies_one$genre, "Other")

table(movies_one$genre2)
```

```
##
##      Comedy      Drama      Other Thriller
##        104        244         92        65
```

The new variable genre2 keeps Drama, Comedy, and Thriller as separate categories, while all other genres (Action, Documentary, Family, Adventure, Horror, Fiction, Fantasy, and any NA genres) are grouped into “Other”. The frequency table shows: Drama (244), Comedy (104), Other (92), and Thriller (65).

step c

```
#WRITE YOUR CODE HERE
movies_one$genre2 <- relevel(factor(movies_one$genre2), ref = "Other")

model_genre <- lm(log_revenue ~ log_budget + genre2 + release_year, data = movies_one)
summary(model_genre)
```

```
##
## Call:
## lm(formula = log_revenue ~ log_budget + genre2 + release_year,
##     data = movies_one)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.5135  -0.5053   0.2392   0.7900   4.7458
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.2764455  27.8518730   0.046   0.963
## log_budget    0.9110155   0.0633968  14.370 <2e-16 ***
## genre2Comedy   0.3678607   0.2813770   1.307   0.192
## genre2Drama   -0.2810763   0.2498695  -1.125   0.261
## genre2Thriller 0.1622609   0.3076178   0.527   0.598
## release_year   0.0004446   0.0138670   0.032   0.974
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.613 on 318 degrees of freedom
## (181 observations deleted due to missingness)
## Multiple R-squared:  0.4122, Adjusted R-squared:  0.403
## F-statistic: 44.6 on 5 and 318 DF, p-value: < 2.2e-16
```

-Intercept (1.28): the predicted log revenue for an “Other”-genre movie, with a log budget and release year of zero, this value has no direct real-world interpretation on its own. -log_budget (0.911): holding genre

and release year constant, a 1% increase in budget is associated with an approximately 0.91% increase in revenue (elasticity interpretation). This effect is highly significant ($p < 0.001$). -genre2Drama (-0.281): holding budget and release year constant, Drama movies are predicted to have about 28% lower revenue than “Other” genre movies, although this difference is not statistically significant ($p = 0.261$). -genre2Comedy (0.368): holding budget and release year constant, Comedy movies are predicted to have about 37% higher revenue than “Other” genre movies, but this is also not statistically significant ($p = 0.192$). -genre2Thriller (0.162): holding budget and release year constant, Thriller movies are predicted to have about 16% higher revenue than “Other” genre movies, not statistically significant ($p = 0.598$). -release_year (0.0004): holding budget and genre constant, each additional year is associated with an essentially negligible (and statistically insignificant, $p = 0.974$) change in log revenue, suggesting no meaningful time trend once budget and genre are controlled for.

step d

```
#WRITE YOUR CODE HERE
model_predict <- lm(log_revenue ~ log_budget + genre2 + release_year + runtime, data =
  ↪ movies_one)
summary(model_predict)
```

```
##
## Call:
## lm(formula = log_revenue ~ log_budget + genre2 + release_year +
##     runtime, data = movies_one)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.2451  -0.4664   0.2219   0.7775   4.6182
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -3.474499   27.628408  -0.126  0.90000
## log_budget    0.835383    0.068606  12.177 < 2e-16 ***
## genre2Comedy   0.539170    0.285546   1.888  0.05991 .
## genre2Drama   -0.370662    0.249540  -1.485  0.13844
## genre2Thriller 0.207607    0.304996   0.681  0.49657
## release_year   0.002641    0.013752   0.192  0.84784
## runtime        0.014696    0.005383   2.730  0.00669 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.596 on 317 degrees of freedom
## (181 observations deleted due to missingness)
## Multiple R-squared:  0.4257, Adjusted R-squared:  0.4148
## F-statistic: 39.16 on 6 and 317 DF, p-value: < 2.2e-16
```

```
new_movie <- data.frame(
  log_budget = log(20000000),
  genre2 = factor("Drama", levels = levels(movies_one$genre2)),
  release_year = 2026,
  runtime = 120
)

pred <- predict(model_predict, newdata = new_movie, interval = "prediction", level =
  ↪ 0.90)
```

```
pred
```

```
##           fit      lwr      upr
## 1 17.31257 14.63197 19.99316
```

```
exp(pred)
```

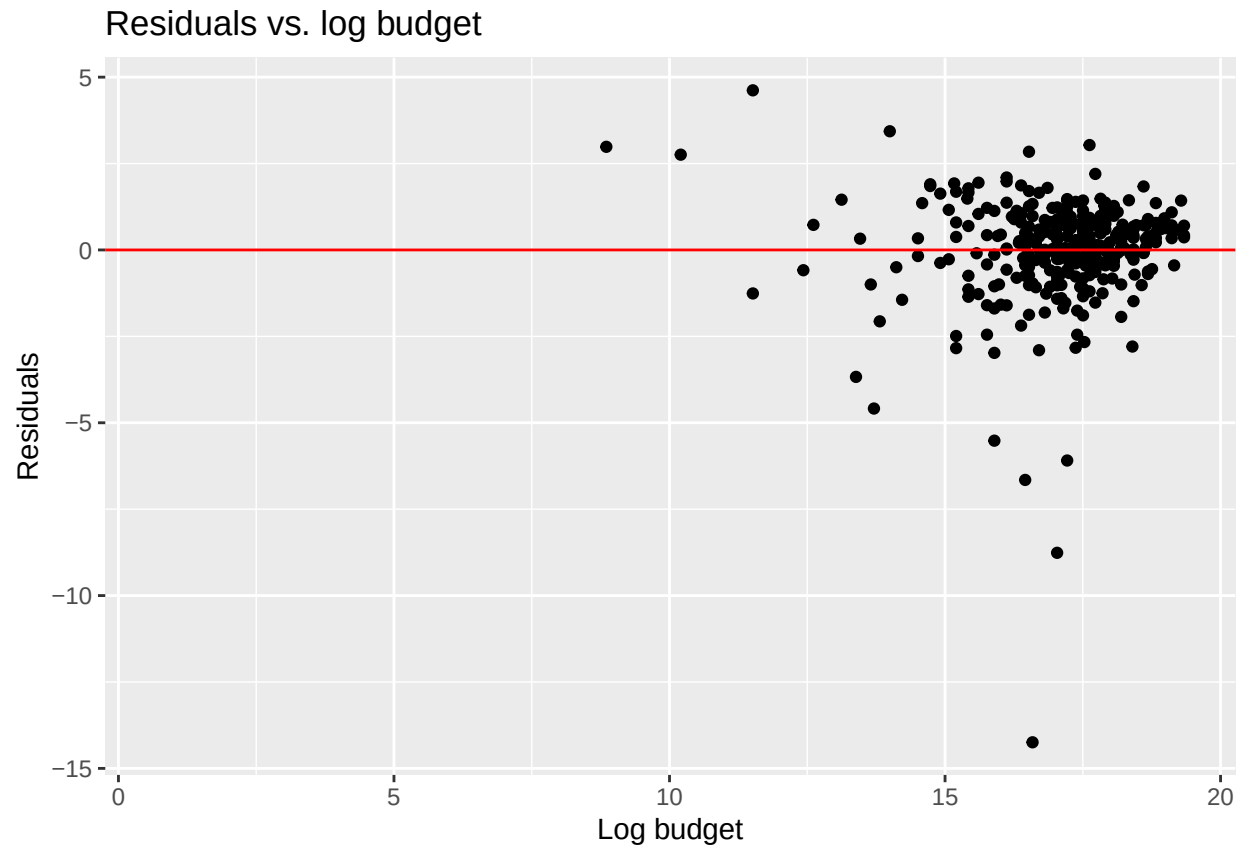
```
##           fit      lwr      upr
## 1 33018130 2262475 481860391
```

The model predicts a log revenue of approximately 17.31 for this movie, which corresponds to a predicted revenue of approximately 33.0 million dollars. The 90% prediction interval for log revenue translates to a revenue interval of roughly 2.3 million dollars to 481.9 million dollars. This is an extremely wide interval, reflecting the large amount of unexplained variation in movie revenue even after accounting for budget, genre, release year, and run time. *step e*

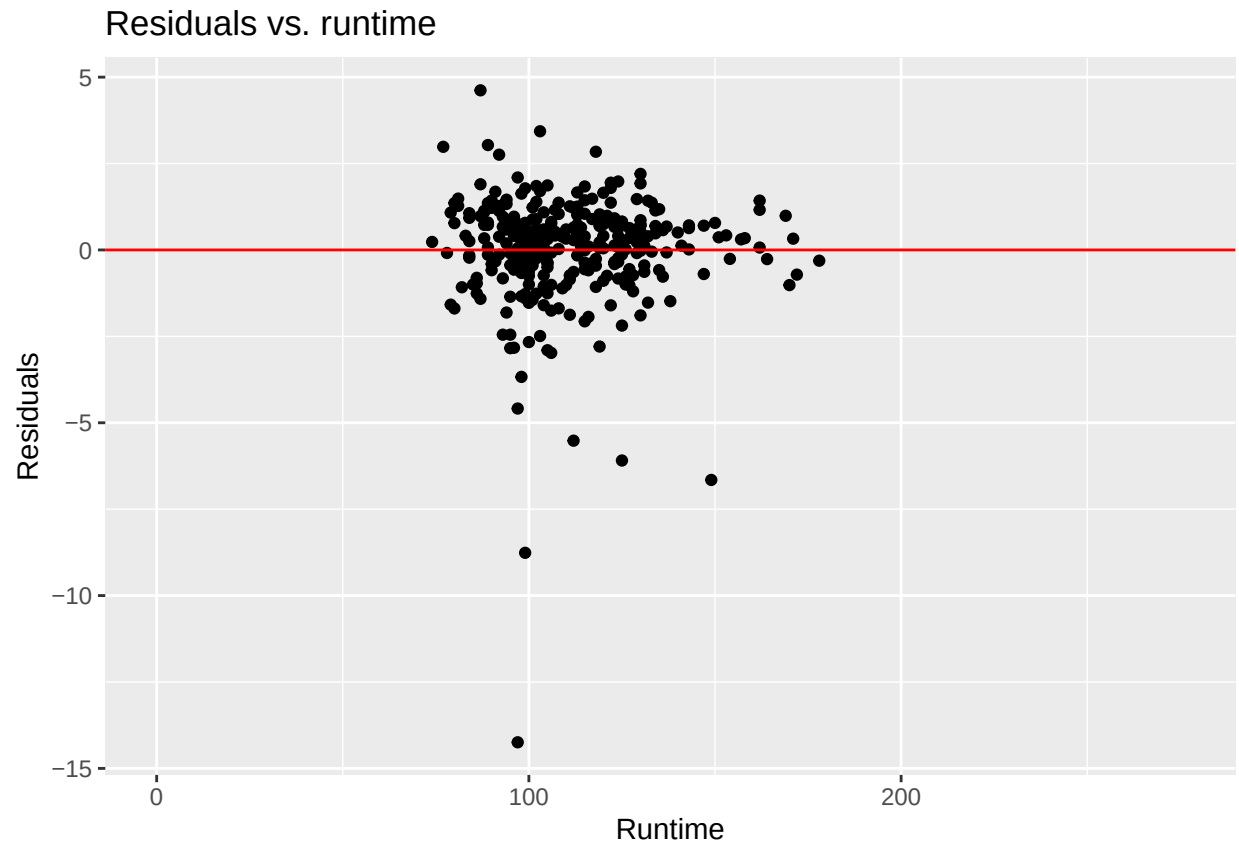
```
#WRITE YOUR CODE HERE
model_predict <- lm(log_revenue ~ log_budget + genre2 + release_year + runtime,
                    data = movies_one, na.action = na.exclude)

movies_one$residuals <- resid(model_predict)
movies_one$fitted <- fitted(model_predict)

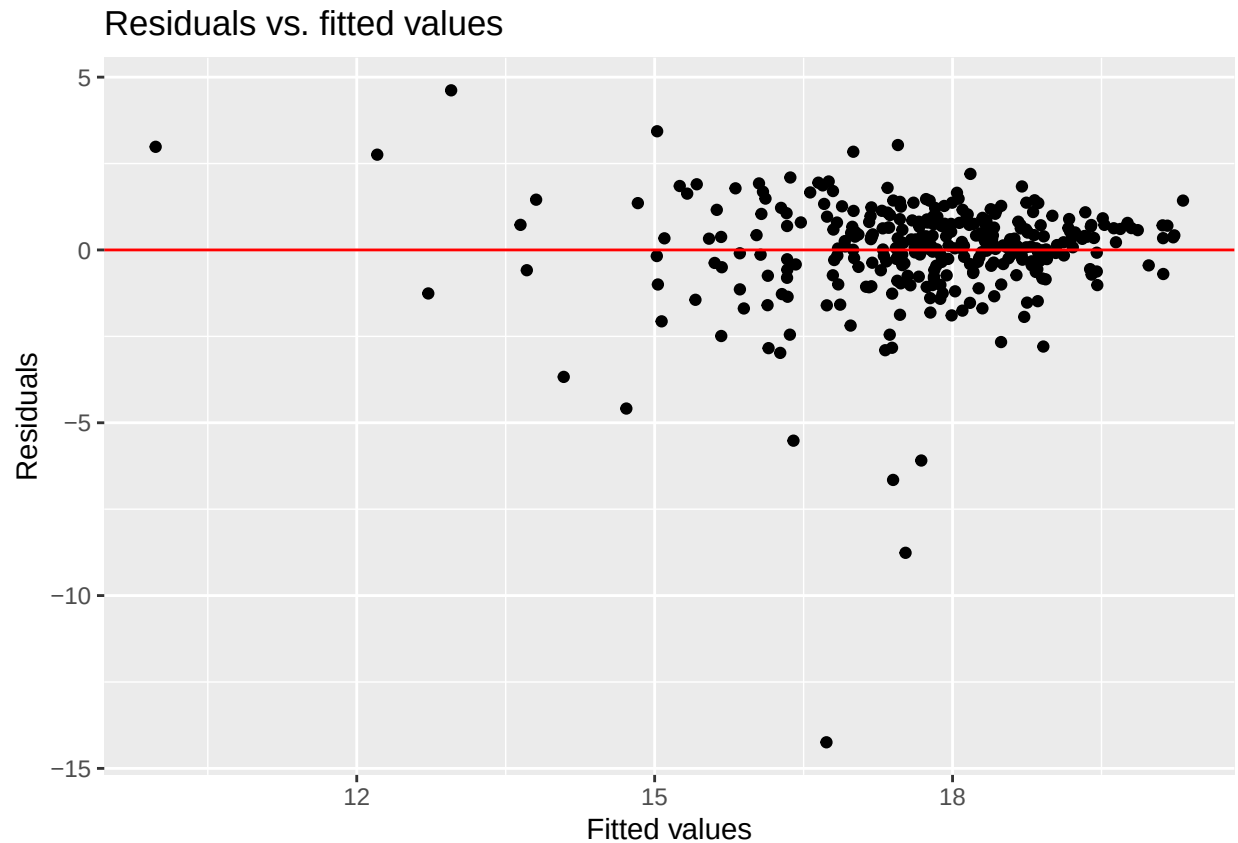
ggplot(movies_one, aes(x = log_budget, y = residuals)) +
  geom_point(na.rm = TRUE) +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Residuals vs. log budget", x = "Log budget", y = "Residuals")
```



```
ggplot(movies_one, aes(x = runtime, y = residuals)) +  
  geom_point(na.rm = TRUE) +  
  geom_hline(yintercept = 0, color = "red") +  
  labs(title = "Residuals vs. runtime", x = "Runtime", y = "Residuals")
```



```
ggplot(movies_one, aes(x = fitted, y = residuals)) +  
  geom_point(na.rm = TRUE) +  
  geom_hline(yintercept = 0, color = "red") +  
  labs(title = "Residuals vs. fitted values", x = "Fitted values", y = "Residuals")
```



The residual plots against log budget, run time, and the fitted values do not show a strong, obvious funnel-shaped pattern, the spread of residuals looks relatively similar across the range of each variable. This suggests that there is not strong evidence of heteroskedasticity in this log-log specification. This makes sense: one of the main benefits of using the log transformation for revenue and budget is that it tends to stabilize the variance compared to using the raw (level) variables, where we would expect much more clear heteroskedasticity.

step f

```
#WRITE YOUR CODE HERE
model_restricted <- lm(log_revenue ~ log_budget + genre2 + release_year, data =
  ↪ movies_one)
model_full <- lm(log_revenue ~ log_budget + genre2 + release_year +
  ↪ factor(release_month), data = movies_one)

anova(model_restricted, model_full)

## Analysis of Variance Table
##
## Model 1: log_revenue ~ log_budget + genre2 + release_year
## Model 2: log_revenue ~ log_budget + genre2 + release_year + factor(release_month)
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     318 826.96
## 2     307 792.33 11    34.629 1.2198 0.2727
```

Null hypothesis (H_0): the coefficients on all the release-month dummies are jointly equal to zero (month of

release has no effect on revenue, after controlling for budget, genre, and release year). Alternative hypothesis (H_1): at least one release-month dummy coefficient is different from zero. Significance level: $\alpha = 0.05$.

Using an F-test comparing the restricted model (without month) to the full model (with month dummies), the F-statistic is approximately 1.22, with a corresponding p-value of approximately 0.27. Since the p-value is much greater than $\alpha = 0.05$, we fail to reject the null hypothesis. There is no statistically significant evidence that the month of release affects movie revenue, once budget, genre, and release year are accounted for.

Advice to the studio: based on this analysis, we would advise the studio that the timing of the release within the year does not appear to have a statistically significant impact on revenue. Rather than focusing on choosing a specific “optimal” release month, the studio would likely get more value from focusing on factors that do significantly predict revenue, such as production budget. That said, this result is purely statistical and based on a relatively small and selective sample; the studio may still want to consider other practical factors (such as competition from other major releases, or qualifying windows for awards season) when timing a release.